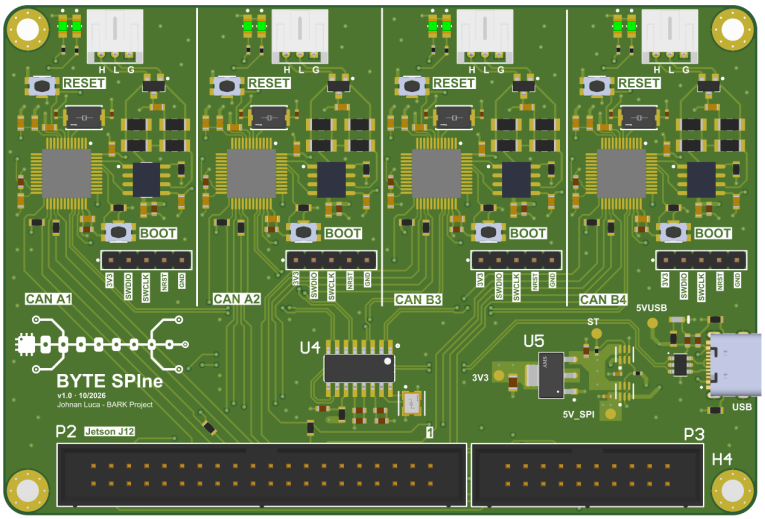


Architecture Design and Sizing of the "BYTE SPIne" SPI-to-CAN Bridge

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A practical documentation for understanding the BYTE SPIne architecture and the sizing of its elements. The impedance and signal integrity simulations are documented in the companion report *Impedance and Signal Integrity Simulation of the BYTE SPIne* (`BYTE_SPIne_SI_Simulation.pdf`).

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1 Abstract and System Overview

The BYTE SPIne is the communication backbone between the high-level computer of the BARK quadruped robot, an NVIDIA Jetson Orin Nano, and its twelve joint actuators. Each leg carries three SteadyWin GIM8108-8 brushless actuators, which are commanded over a classic CAN 2.0 bus (standard 11-bit frames, 8-byte payload, up to 1 Mbit/s). The board creates **four independent CAN buses**, one per leg, each driven by its own STM32F072C8T6 microcontroller and TCAN1042HV transceiver.

The board is designed to be operated in two modes on the same hardware:

- **USB mode (development):** the four MCUs are connected to the Jetson through an on-board four-port USB hub (SL2.1A). Each MCU runs the open-source *candleLight* firmware and appears in Linux as a standard `gs_usb` SocketCAN interface (`can0`–`can3`). This mode allows the robot to be operated immediately, while the SPI firmware is being developed.
- **SPI mode (final):** the four MCUs act as SPI slaves on the two SPI controllers of the Jetson 40-pin expansion header. The MCU runs the time-critical CAN loop locally, while the Jetson only exchanges setpoints and motor states.

Supporting both modes on one design avoids a second PCB order when moving from USB to SPI, and makes use of the minimum order quantity of the PCBA service (JLCPCB). All SMD components are chosen from the JLCPCB assembly catalogue, with priority to *Basic* parts (no setup fee) [12]; this choice drives several sizing decisions in the following sections. The through-hole connectors are soldered by hand.

The power of the board is drawn from the Jetson itself, either from the USB VBUS or from the 5 V pins of the expansion header, through an ideal-diode OR-ing stage.

1.1 Requirements for Correct Use

The following points are mandatory to operate the board correctly. They are explained in the referenced sections and highlighted there in red boxes.

Wiring and setup

1. The GND of the SPIne must be common with the ground of the motor power supply (Section 4.4).
2. On each leg, the CAN termination must be enabled **only on the last actuator**; all actuators must run at 1 Mbit/s (Sections 4.2 and 4.3).
3. Ribbon cables are connected and disconnected only with the robot switched off (Section 5.3).
4. At most 0.1 A may be drawn from pin 1 of P3, and no further loads may be added to the 5 V rail (Sections 5.2 and 7.3).

Jetson

5. The SPI controllers must be enabled once with `jetson-io`; SPI clock 1 MHz to 4 MHz, ribbon cable shorter than 15 cm to 20 cm (Section 5.4).

Firmware (SPI mode)

6. Clock from the 8 MHz crystal (HSE), Serial Wire debug enabled, SPI and SPI_INT pins at low output speed (Section 5.7).
7. An MCU whose NSS is high must release MISO (high impedance); to be verified on the bench (Section 5.5).
8. CRC, sequence counter and communication watchdog must be implemented before the robot is operated in SPI mode (Section 5.6).

2 System Architecture

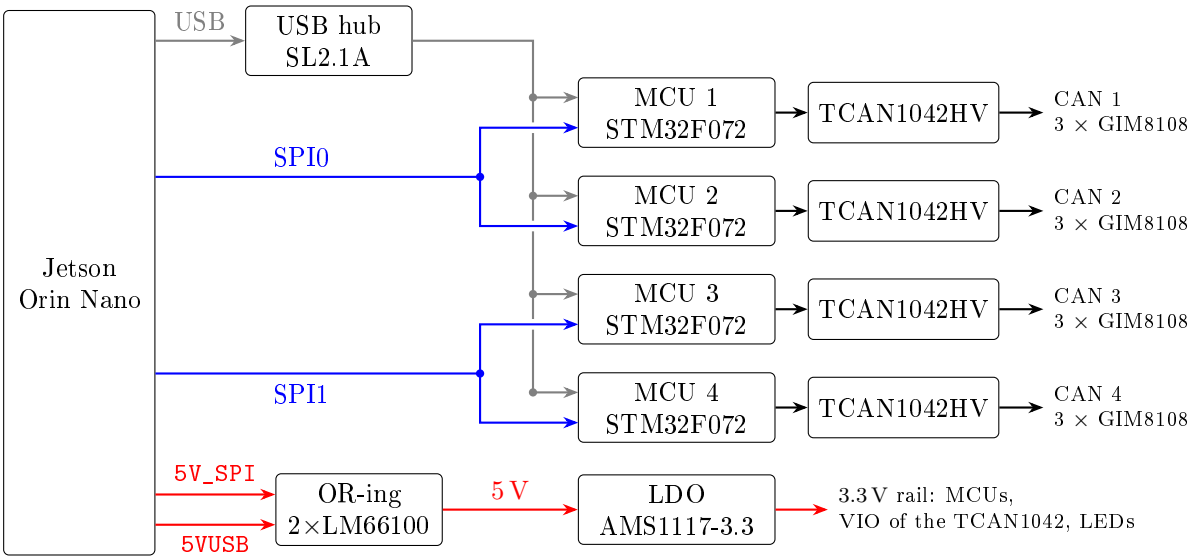


Figure 1: Block diagram of the BYTE SPIne. SPI0 serves MCU 1–2, SPI1 serves MCU 3–4; the USB hub provides the development path (candleLight), with one downstream port per MCU. The board is supplied by the two 5 V sources of the Jetson (`5V_SPI` from the 40-pin header, `5VUSB` from the USB VBUS), combined by the ideal-diode OR-ing into the 5 V rail (transceivers, USB hub, LDO). The LDO generates the 3.3 V rail.

The schematic is hierarchical. The sheet `CAN_channel.SchDoc` describes one complete channel (MCU, clock, debug, transceiver, termination and protection) and is instantiated four times through two multi-channel sheet symbols, `Repeat(CAN_A,1,2)` and `Repeat(CAN_B,3,4)`. Grouping the channels in pairs mirrors the two SPI buses of the Jetson and keeps the net names unique (`SPI0_*` and `SPI1_*`). Component designators therefore carry a channel suffix, e.g. `U1_CAN_A1` or `C3_CAN_B4`.

2.1 MCU Pin Assignment

STM32F072 pin	Function	Notes
PA0, PA1	Status LEDs (green, yellow)	driven through R1, R2
PA11, PA12	USB DM, DP	to the hub downstream port
PA13, PA14	SWDIO, SWCLK	debug header P1
PB8, PB9	CAN_RX, CAN_TX	AF4, bxCAN
PB11	SPI_INT (GPIO output)	data-ready line to the Jetson
PB12–PB15	SPI2 NSS, SCK, MISO, MOSI	SPI slave
PF0, PF1	OSC_IN, OSC_OUT	8 MHz crystal
BOOT0	Boot selection	button S1, pull-down R10
NRST	Reset	button S2, C4

Table 1: Pin assignment of each MCU channel.

3 Microcontroller Subsystem (STM32F072C8T6)

3.1 Clock Source: Why an External Crystal

The STM32F072 is a crystal-less USB device: in USB mode its internal 48 MHz oscillator (HSI48) is trimmed continuously by the Clock Recovery System using the USB start-of-frame packets. This is not sufficient for CAN. When the board runs in SPI mode, no USB host is present and the internal oscillators are untrimmed. The datasheet [1] specifies:

- HSI (8 MHz): ±1 % at 25 °C, −2.8/ + 3.8 % over −40 °C to 105 °C;
- HSI48 (factory calibrated): −2.8/ + 2.9 % at 25 °C, −4.9/ + 4.7 % over the full range.

A CAN network at 1 Mbit/s typically tolerates an oscillator error well below 1 %. An 8 MHz crystal (YXC X50328MSB2GI, ±10 ppm, $C_L = 20\,\text{pF}$) is therefore mounted on PF0/PF1. The crystal also feeds the PLL for USB, so both modes work from the same clock tree.

3.2 Crystal Load Capacitors

Following AN2867 [13], the two load capacitors appear in series as seen by the crystal, in parallel with the stray capacitance of pins and traces ($C_{stray} \approx 3\,\text{pF}$ to 5 pF):

$$C_L = \frac{C_7\,C_8}{C_7 + C_8} + C_{stray} \quad \Rightarrow \quad C_7 = C_8 = 2\,(C_L - C_{stray}) = 2\,(20 - 3.5)\,\text{pF} \approx 33\,\text{pF} \tag{1}$$

With the selected 33 pF C0G capacitors the effective load is $16.5 + (3 \ldots 5) = 19.5\,\text{pF}$ to 21.5 pF, centred on the 20 pF specification of the crystal.

3.3 Power Supply Decoupling

The decoupling follows the power supply scheme of the datasheet [1] (figure 13) and of AN4080 [14]: one 100 nF capacitor per VDD pin plus one common 4.7 µF, a dedicated 100 nF + 4.7 µF pair on VDDIO2, and 10 nF + 1 µF on VDDA. All capacitors are on the 3.3 V net, so their assignment to a pin is defined by the placement:

- VDD pin 24: C14 (100 nF) and C13 (4.7 µF, common VDD bulk);
- VDD pin 48 (and VBAT, pin 1, at the same corner): C10 (100 nF);
- VDDIO2 pin 36: C15 (100 nF) and C5 (4.7 µF);
- VDDA pin 9: C11 (10 nF, closest to the pin) and C12 (1 µF).

The 100 nF capacitors must sit directly at their pins, because they supply the fast current edges. The 4.7 µF bulk capacitors cover slow load changes and are much less sensitive to their position: they reach the other pins through the 3.3 V and GND planes, whose inductance is about 1 nH (6 mΩ at 1 MHz).

3.4 Boot, Reset and Debug

BOOT0 is held low by R10 (10 kΩ); pressing S1 at reset pulls it to 3.3 V and starts the ST system bootloader (USB DFU), which is the standard way to flash candleLight. NRST is filtered by C4 (100 nF) against the internal pull-up of about 40 kΩ; S2 provides a manual reset. The frame terminal (pin 5) of both KMR221G buttons is tied to GND for ESD.

Each channel exposes a 5-pin SWD header P1 (3V3, SWDIO, SWCLK, NRST, GND) for programming and debugging with an ST-Link, independently of USB. The open-drain reset output of the probe is compatible with C4.

4 CAN Physical Layer

4.1 Transceiver: TCAN1042HV

The TCAN1042HVDR is supplied with $V_{CC} = 5\,\mathrm{V}$ (recommended range 4.5 V to 5.5 V) and uses the VIO pin at 3.3 V (range 3.0 V to 5.5 V) to match the logic levels of the STM32 without external level shifters. Its bus pins withstand faults up to ±58 V, which is relevant in a robot where motor supply lines run next to the CAN wiring. The STB pin is tied to GND through R5 (10 kΩ) so the device is always in normal mode; the internal pull-up of STB draws at most 20 µA, giving less than 0.2 V on the pin.

Decoupling: 4.7 µF + 100 nF on VCC (C1, C2), 100 nF on VIO (C6).

4.2 Bus Topology and Bus Load

Each bus connects the SPIne (first node) to the three actuators of one leg. The GIM8108-8 drivers [9] use standard 11-bit frames with an 8-byte payload in a request/response scheme, i.e. two frames per motor per control cycle. A standard frame with 8 data bytes is 111 bits long without bit stuffing and up to about 135 bits in the worst case:

$$t_{frame} = \frac{111 \ldots 135\ \mathrm{bit}}{1\,\mathrm{Mbit/s}} = 111\,\mu\mathrm{s}\ \text{to}\ 135\,\mu\mathrm{s} \tag{2}$$

$$t_{cycle} = 3\ \text{motors} \times 2\ \text{frames} \times t_{frame} = 0.67\,\mathrm{ms}\ \text{to}\ 0.81\,\mathrm{ms} \tag{3}$$

At a 1 kHz control rate the bus load is therefore 67 % to 81 %, at 500 Hz it is 33 % to 41 %. A 1 kHz loop is feasible only because each leg has its own bus, and requires all actuators to be configured at 1 Mbit/s.

Required: configure **all actuators of the bus at** 1 Mbit/s.

4.3 Split Termination Sizing

A CAN bus must be terminated with 120 Ω at both physical ends. Since the SPIne is always the first node of each bus, the termination is placed on the board; the last actuator of each leg must provide the other one, and the intermediate actuators must have their termination disabled.

Required: on each leg, enable the CAN termination **only on the last actuator** of the bus and disable it on the intermediate ones.

The termination is split in two halves of 60 Ω with the centre tap connected to GND through $C_{split} = 4.7\,\mathrm{nF}$ (C3). This stabilises the common-mode voltage and reduces emissions. Each half is built from **two 120 Ω 1206 resistors in parallel** (R6/R15 and R7/R16):

$$R_{half} = \frac{120\,\Omega}{2} = 60\,\Omega, \qquad R_{term} = 2\,R_{half} = 120\,\Omega \tag{4}$$

Why four identical resistors. At first sight four resistors, two in parallel and the two pairs in series, look unnecessary: one 60 Ω resistor per half would give the same 120 Ω. The reason is cost. Each half must withstand the fault power computed below (0.6 W peak), which requires a large package (2010 or 2512) for a single resistor. In the JLCPCB assembly catalogue [12] such resistors are only available as *Extended* parts, which add a setup fee per part type and a higher unit price. A 120 Ω 1206 resistor (0.25 W) is instead a *Basic* part, without setup fee. Two of them in parallel give exactly 60 Ω and double the power rating to 0.5 W per half. The price is one extra small resistor per half, which costs much less than an Extended part.

Normal operation. The dominant differential voltage is at most $V_{OD(DOM)} = 3\,\mathrm{V}$ (TCAN1042 datasheet [2], $R_L = 50\,\Omega$ to $65\,\Omega$), i.e. 1.5 V on each half:

$$P_{half} = \frac{(1.5\,\mathrm{V})^2}{60\,\Omega} = 37.5\,\mathrm{mW}, \qquad P_{resistor} = \frac{P_{half}}{2} \approx 19\,\mathrm{mW} \tag{5}$$

This is a peak value during dominant bits; the average is lower.

Fault condition. The datasheet [2] (section 9.3.6) requires the termination to be sized for bus fault cases. If a bus line is shorted to a supply rail while the node transmits a dominant bit, the current is limited by the driver to $I_{OS(SS_DOM)} \leq 100\,\mathrm{mA}$:

$$P_{half,fault} = I_{OS}^2 \times R_{half} = (0.1\,\mathrm{A})^2 \times 60\,\Omega = 0.6\,\mathrm{W}, \qquad P_{resistor,fault} = 0.3\,\mathrm{W} \tag{6}$$

This is a peak value: the average is reduced by the ratio of recessive bits and by the TXD dominant time-out of the transceiver. A single 60.4 Ω 0603 resistor (100 mW) would not survive this case; the chosen arrangement provides 0.5 W per half using only JLCPCB Basic parts (`C17909`).

4.4 ESD Protection

A Nexperia PESD1CAN (D1) [6] protects both lines: pins 1 and 2 (cathodes) are connected to CANH and CANL, pin 3 (common) to GND. The connector J1 is a JST PH 3-pin (S3B-PH-K-S) with the pinout 1 = GND, 2 = CANL, 3 = CANH.

Required: the GND of the SPIne must be **common with the ground of the motor power supply**; without a common reference the CAN common-mode range can be exceeded.

5 SPI Interface to the Jetson Orin Nano

5.1 Bus Mapping

The Jetson Orin Nano developer kit exposes two SPI controllers on the 40-pin header J12 [7, 11], each with two hardware chip selects. The four MCUs are therefore split in two pairs, so that each MCU has a dedicated hardware chip select and each MISO line is shared by only two slaves.

Channel	Jetson bus	SCK / MOSI / MISO (pin)	NSS (pin)	INT (pin)
CAN_A1	SPI0	23 / 19 / 21	24 (CS0)	29
CAN_A2	SPI0	23 / 19 / 21	26 (CS1)	31
CAN_B3	SPI1	13 / 37 / 22	18 (CS0)	32
CAN_B4	SPI1	13 / 37 / 22	16 (CS1)	33

Table 2: Mapping between the MCU channels and the Jetson 40-pin header (P2). 5 V on pins 2 and 4; GND on pins 6, 9, 14, 20, 25, 30, 34, 39. The 3.3 V pins 1 and 17 (`3V3_Jetson`) are not connected to the 3.3 V rail of the board: they are only routed to the breakout header P3.

The connection uses a shrouded 2×20 box header (P2) and a 40-way IDC ribbon cable, so that the cable cannot be inserted reversed.

5.2 Breakout Header P3

The SPIne occupies most of the Jetson header. To keep the remaining Jetson signals usable (sensors, debug UART, I2S), they are routed one-to-one to a 2×10 shrouded header P3 (Würth 61202021621):

Pin	Signal (J12 pin)	Pin	Signal (J12 pin)
1	3V3_Jetson (1, 17)	2	GND
3	I2C1_SDA (3)	4	I2C1_SCL (5)
5	I2C0_SDA (27)	6	I2C0_SCL (28)
7	GND	8	GND
9	UART1_TXD (8)	10	UART1_RXD (10)
11	UART1_RTS (11)	12	UART1_CTS (36)
13	GND	14	GPIO09 (7)
15	GPIO12 / PWM (15)	16	I2S0_SCLK (12)
17	I2S0_FS (35)	18	I2S0_DIN (38)
19	I2S0_DOUT (40)	20	GND

Table 3: Pinout of the breakout header P3. The 5 V rail is not exported on P3: the 0.5 A budget of the Jetson header is reserved for the SPIne.

Required: P3 pin 1 (`3V3_Jetson`) is limited to 0.1 A by the Jetson; sensors connected to P3 must stay within this budget or be supplied separately.

5.3 ESD Protection of the Jetson Connectors

ESD protection is placed only on the connectors exposed to handling or to external cables: the USB-C port (USBLC6-2SC6) and the four CAN connectors (PESD1CAN). P2 and P3 are not protected on purpose:

- P2 is an internal board-to-board connection to the Jetson through a short ribbon cable inside the robot, connected only during assembly; the header signals already pass through the TXB0108 translators of the carrier board;
- a TVS array on the SPI lines would add capacitance to signals that are already at about 40 pF, while the TXB0108 is sensitive to capacitive loads;
- P3 is intended for sensors inside the robot with short cables. If it is ever used with long or external cables, the protection should be placed on the sensor side or on an adapter, or added in a later revision.

Required: connect and disconnect the ribbon cables **only with the robot switched off**, and handle the boards with ESD precautions.

5.4 Level Translators on the Jetson Side

According to the carrier board specification (SP-11324-001) [7], all header signals are 3.3 V and pass through TI TXB0108 auto-direction translators [8], whose output drivers are very weak. NVIDIA requires any pull-up or pull-down on these signals to be weaker than 50 kΩ. For this reason:

- the NSS pull-up R3 is 100 kΩ (not the usual 10 kΩ). It keeps each slave deselected when the cable is unplugged (USB mode) or while the Jetson boots;
- the internal pull-up of the STM32 (25 kΩ to 55 kΩ) is not used for NSS, as it could fall below the limit;
- the SPI clock should be kept in the range 1 MHz to 4 MHz and the ribbon cable short (below 15 cm to 20 cm).

Required: the SPI pins of the Jetson are configured as GPIO by default: **enable the two SPI controllers once with** `jetson-io` (`sudo /opt/nvidia/jetson-io/jetson-io.py`, then reboot). Keep the SPI clock in the range 1 MHz to 4 MHz and the ribbon cable shorter than 15 cm to 20 cm.

5.5 Data Rate and Shared MISO

Each MCU exchanges roughly 100 bytes per millisecond (three motors, command and state), i.e. below 1 Mbit/s; a 2–4 MHz SPI clock leaves a wide margin. A series resistor R4 (33 Ω) on MISO limits the current in case two slaves drive the line at the same time; placed at the MCU pin, it also works as a source termination of the line (Section 9.5).

Series resistors on SCK. The two SCK lines also have a 33 Ω series resistor, R17 (SPI1_SCK) and R18 (SPI0_SCK), placed at the Jetson side of the line, next to P2. The TXB0108 translators of the Jetson drive very fast edges (0.6 ns to 2.5 ns), so the clock line of each bus, which branches to two MCUs, behaves as a transmission line: every edge is reflected at the ends of the branches and returns as ringing. Without the resistor, the first ring after the rising edge falls below the guaranteed V_{IH} of the STM32, so the clock level is no longer guaranteed to be read correctly during the ring. The series resistor, together with the low output impedance of the driver, matches the line at the source and damps these reflections. This was found with the signal integrity simulation of the routed board, which also gives the value of the resistor (SI report [17], Section 3.8); the results are summarised in Section 9.5.

Required: the SPI firmware must guarantee that an MCU whose NSS is high **releases MISO** (high impedance). This must be verified on the bench with two MCUs on the same bus before the full system test.

5.6 Recommended Firmware Architecture

The Jetson runs a standard (non real-time) Linux kernel, so SPI transfers can occasionally be delayed by milliseconds. The design assumes that:

- the MCU owns the fast CAN loop, timed by its own timer, and keeps using the last valid setpoint if a packet is late;
- SPI packets have a fixed length, a hardware CRC and a sequence counter, and are handled with circular DMA;
- a communication watchdog puts the motors in a safe state (zero torque or damping) if no valid packet arrives within a few milliseconds;
- the PREEMPT_RT kernel available for JetPack is used on the Jetson for the final system.

5.7 MCU Peripheral Configuration (STM32CubeMX)

The SPI-mode firmware must configure the peripherals explicitly; the pins of the SPI2 controller are plain GPIOs after reset. This configuration is not needed in USB mode, where candleLight provides its own board configuration.

Peripheral	Configuration
RCC (clock)	HSE 8 MHz crystal (PF0 = OSC_IN, PF1 = OSC_OUT) → PLL ×6 → SYSCLK = HCLK = PCLK = 48 MHz. The HSE is mandatory for an accurate CAN bit rate (Section 3.1).
SPI2	Mode *Full-Duplex Slave*, *Hardware NSS Input Signal* (on PB12); 8-bit data, MSB first; CPOL/CPHA identical to the Jetson (spidev default: mode 0, CPOL = 0, CPHA = 0); RX and TX on DMA in circular mode (CubeMX assigns SPI2_RX/TX to DMA1 channels 4/5).
SPI2 pins	PB12 = NSS, PB13 = SCK, PB14 = MISO, PB15 = MOSI; alternate function AF0, no internal pull (R3 provides the external 100 kΩ pull-up on NSS), **low output speed**.
Packet integrity	CRC computed on each packet with the CRC peripheral of the STM32F072 (programmable polynomial), so that the Jetson can compute the same value in software. The built-in SPI CRC is an alternative, but with circular DMA it requires the Jetson to append a matching CRC at the end of every frame.
bxCAN	PB8 = CAN_RX, PB9 = CAN_TX; alternate function AF4. 1 Mbit/s from PCLK = 48 MHz: prescaler 3 (t_q = 62.5 ns), SJW = 1, BS1 = 13 t_q, BS2 = 2 t_q, i.e. 16 t_q per bit and a sample point of 87.5 %.
GPIO	PB11 = SPI_INT: push-pull output, low speed, initialised to the inactive level. PA0 = green LED, PA1 = yellow LED: push-pull outputs.
SYS	Debug *Serial Wire* enabled (PA13 = SWDIO, PA14 = SWCLK), otherwise CubeMX may release the pins and the SWD probe loses access.
IWDG	Independent watchdog (LSI) for MCU lock-ups, in addition to the communication watchdog implemented in software.

Table 4: STM32CubeMX configuration of each MCU for the SPI-mode firmware.

Required: in the SPI firmware, the clock must come from the **HSE crystal**, the debug interface must stay enabled (**Serial Wire**) and the SPI and SPI_INT pins must be at **low output speed**.

Hardware note: in slave mode the MISO output must be in high impedance while NSS is high, otherwise the two MCUs of a bus drive MISO at the same time. This is the bench test listed in the next steps (Section 11); if the peripheral does not release the pin, the firmware must switch PB14 between alternate function and input on the NSS edges (EXTI on PB12).

6 USB Subsystem

6.1 USB Type-C Upstream Port

J2 (TYPE-C-31-M-12) is configured as a USB 2.0 device (UFP): CC1 and CC2 are pulled down with 5.1 kΩ (R11, R12) so that the host enables VBUS. D+ and D− are protected by a USBLC6-2SC6 (U3) placed close to the connector. VBUS is decoupled with 10 µF + 100 nF, keeping the total bulk capacitance at the 10 µF limit for a USB device. A red LED (DS1, R13 = 1 kΩ) indicates the presence of VBUS.

6.2 USB Hub: SL2.1A

The SL2.1A is a single-supply USB 2.0 four-port hub with internal 3.3 V and 1.8 V regulators. Following the reference circuit of its datasheet [3]:

- VDD33 (pin 13) and VDD18 (pin 14): 4.7 µF + 100 nF each (C9/C16, C17/C18), placed as close as possible to the pins;

- VCC5V (pin 11): RC filter made of R8 = 4.7 Ω and C19 = 4.7 µF;
- crystal Y2: 12 MHz, $C_L = 20$ pF, ≤ 20 ppm (YXC X322512MSB4SI, ±10 ppm). No load capacitors are required by the SL2.1A.

The four downstream ports are routed directly to the USB pins of the four MCUs.

6.3 candleLight Firmware

In USB mode each STM32F072 runs candleLight (`candle-usb/candleLight_fw`) [10], whose reference hardware uses the same MCU with CAN on PB8/PB9. The Linux `gs_usb` driver exposes each channel as a SocketCAN interface, so no driver needs to be written on the Jetson side.

7 Power Architecture

7.1 Ideal-Diode OR-ing (LM66100)

The board can be powered from two 5 V sources of the Jetson: `5V_SPI` (header pins 2 and 4) and `5VUSB` (USB VBUS). A Schottky OR-ing would drop 0.4 V to 0.5 V at the board current; starting from a USB VBUS of 4.75 V, this would leave about 4.3 V, below the 4.5 V minimum of the TCAN1042 (whose VCC undervoltage protection rises at 4.2 V to 4.4 V). Two TI LM66100 ideal diodes are used instead:

$$V_{drop} = I \times R_{ON} = 0.43\,\text{A} \times 79\,\text{m}\Omega \approx 34\,\text{mV}, \qquad P = I^2 R_{ON} \approx 15\,\text{mW} \tag{7}$$

Since both sources come from the same internal rail of the Jetson (VDD_5V_SYS), their voltages are almost identical. The simple cross-coupled OR-ing (LM66100 datasheet [5], figure 15) can switch both devices off in this condition. The circuit therefore implements figure 17, "Dual Ideal Diode ORing for Continuous Output Power":

- U6 (priority, `5V_SPI`): CE connected to `5VUSB`; ST pulled up to `5V_SPI` by R9 (100 kΩ) and connected to the CE of U7;
- U7 (secondary, `5VUSB`): CE driven by the ST of U6; ST pulled up to 5 V by R14 and brought to test point TP2.

Scenario	Behaviour
Only `5V_SPI`	U6 on (its CE sees 0 V), ST of U6 high-Z, R9 pulls CE of U7 high: U7 off.
Only `5VUSB`	U6 unpowered, pull-up R9 at 0 V: CE of U7 low, U7 on.
Both present	U6 has priority; if U6 switches off because the inputs are equal, its ST goes low and enables U7 immediately, so the output is never left unpowered.

Table 5: Operating scenarios of the OR-ing stage. TP2 (ST of U7) reads high when U7 is conducting.

Each LM66100 has a 1 µF input capacitor (C24, C25).

7.2 3.3 V Regulator (AMS1117-3.3)

The 3.3 V rail supplies the four MCUs, the VIO pins of the transceivers and the LEDs. With a typical MCU current of 24 mA at 48 MHz (datasheet [1], all peripherals enabled) the load is about 150 mA:

$$P_{LDO} = (V_{IN} - V_{OUT}) \times I = (4.95 - 3.3)\,\text{V} \times 0.15\,\text{A} \approx 0.25\,\text{W} \tag{8}$$

which is easily handled by the SOT-223 package. The dropout (1.1 V at 0.8 A, lower at this current) requires $V_{IN} \gtrsim 4.4\,\text{V}$, which the ideal-diode OR-ing preserves.

Hardware note: the AMS1117 datasheet [4] guarantees stability for all conditions with a 22 µF solid tantalum output capacitor. The design uses a 22 µF 0805 25 V X5R ceramic capacitor (C21), a common and generally stable choice. The 3.3 V rail must be checked for oscillation with an oscilloscope on the prototype; if needed, C21 can be replaced by a tantalum capacitor.

7.3 Current Budget

Load	Current (worst case)
4 × TCAN1042 (dominant, $R_L = 60\,\Omega$, max)	280 mA
3.3 V rail through the LDO (4 MCUs, LEDs, VIO)	150 mA
USB hub and VBUS LED (estimate)	30 mA
Total on the 5 V rail	≈ 0.46 A

Table 6: Worst-case current budget. In normal operation the transceivers draw 2.5 mA in the recessive state, so the typical current is significantly lower.

Required: the Jetson carrier board specification limits the 5 V current to 0.5 A for the 40-pin header and 0.5 A per USB port. The worst case is within this limit with little margin: **no further loads may be added to the 5 V rail**.

8 LEDs

8.1 LED Resistor Sizing

The series resistor of each LED is sized with Ohm's law:

$$R = \frac{V_{rail} - V_f}{I_f} \qquad \Longleftrightarrow \qquad I_f = \frac{V_{rail} - V_f}{R} \tag{9}$$

LED	Part	Rail	V_f	R	I_f
D2 green	KT-0805G	3.3 V (PA0)	2.6 V to 3.1 V	100 Ω	2 mA to 7 mA
D3 yellow	KT-0805Y	3.3 V (PA1)	1.8 V to 2.4 V	1 kΩ	0.9 mA to 1.5 mA
DS1 red	KT-0603R	5 V (VBUS)	1.8 V to 2.4 V	1 kΩ	2.6 mA to 3.2 mA

Table 7: LED operating points. The green LED has a high forward voltage: with 1 kΩ it would be almost off at 3.3 V, hence the 100 Ω resistor.

8.2 BYTE SPIne LEDs Color Code

- **RED (DS1):** USB VBUS present.
- **GREEN (D2, per channel):** MCU status / CAN activity (defined by the firmware).
- **YELLOW (D3, per channel):** MCU status / CAN activity (defined by the firmware).

9 PCB Layout

9.1 Stack-up

The board uses the standard JLCPCB 4-layer stack-up JLC04161H-7628 (1.6 mm) [16]. Signals are routed on the outer layers; the two inner layers are reference and distribution planes.

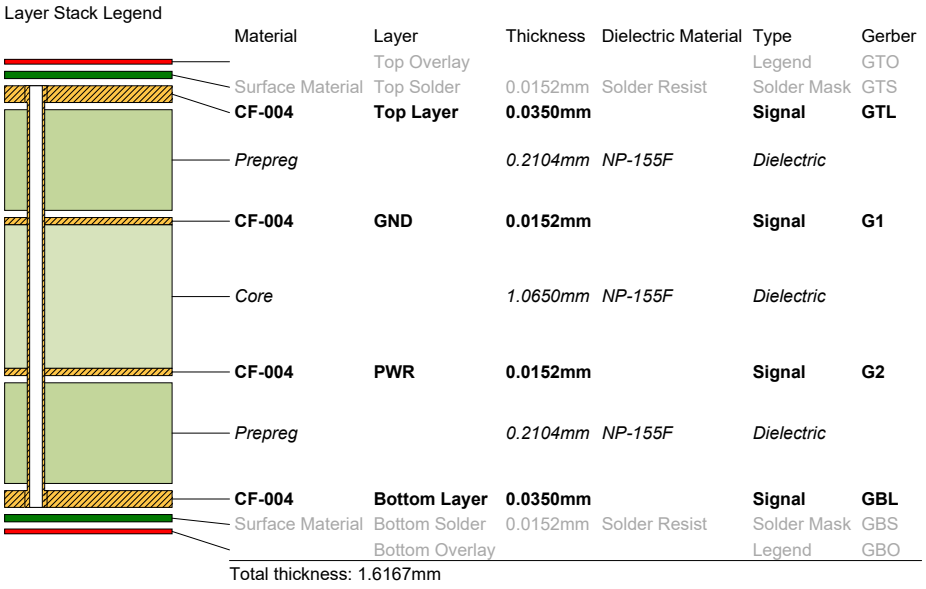
Layer	Function	Copper	Dielectric to next layer
L1 Top	signals, components	35 µm (1 oz)	prepreg 0.2104 mm, $\varepsilon_r = 4.4$
L2 GND	solid GND plane	15.2 µm (0.5 oz)	core 1.065 mm, $\varepsilon_r = 4.43$
L3 PWR	3.3 V plane with 5 V tracks	15.2 µm (0.5 oz)	prepreg 0.2104 mm, $\varepsilon_r = 4.4$
L4 Bottom	crossings, GND pour	35 µm (1 oz)	–

Table 8: Stack-up. Top-layer signals reference the GND plane (L2); bottom-layer signals reference the 3.3 V plane (L3), which is AC-coupled to GND by the decoupling capacitors.

The impedance of a track depends only on its cross-section: track width w and copper thickness t, height h and permittivity ε_r of the dielectric to the reference plane and, for a differential pair, the gap s between the two lines. For an edge-coupled microstrip the closed-form approximation of IPC-2141 gives

$$Z_0 \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\!\left(\frac{5.98\,h}{0.8\,w + t}\right), \qquad Z_{\text{diff}} \approx 2Z_0\left(1 - 0.48\,e^{-0.96\,s/h}\right). \tag{10}$$

With $w = 0.25$ mm, $s = 0.15$ mm, $t = 35$ µm, $h = 0.2104$ mm and $\varepsilon_r = 4.4$ this yields $Z_0 \approx 61\,\Omega$ and $Z_{\text{diff}} \approx 92\,\Omega$, without solder mask. The exact value is obtained with the 2D field solver of the Altium Layer Stack Manager, which solves the electromagnetic field of the real cross-section including the solder mask: $90.18\,\Omega$ (Figure 2). The solder mask lowers the impedance by a few ohms, which explains the 2 % difference; the agreement between the two methods confirms both the geometry of the pairs and the stack-up entered in the design.



Transmission Line Structure Table

Transmission Line	Target Impedance	Calculated Impedance	Trace layer	Width	Gap	Reference layers	Target Tolerance
Edge-Coupled Coated Microstrip	90	90.18	Top Layer	0.2500mm	0.1500mm	GND	10%
Edge-Coupled Offset Stripline	90	90.03	GND	0.1578mm	0.1270mm	Top Layer,PWR	10%
Edge-Coupled Offset Stripline	90	90.03	PWR	0.1578mm	0.1270mm	GND,Bottom Layer	10%
Edge-Coupled Coated Microstrip	90	90.18	Bottom Layer	0.2500mm	0.1500mm	PWR	10%

Figure 2: Layer stack and differential impedance profile (Altium Draftsman). The USB pairs (0.25 mm width, 0.15 mm gap) are routed only on L1 and L4, where they reach 90.18 Ω (target 90 Ω ±10 %). The inner-layer rows belong to the same profile but are not used.

9.2 Board Outline and Floorplan

The board measures about 108×73 mm with four plated M3 mounting holes (3.2 mm drill, 6 mm pad, eight stitching pads) placed 3.5 mm from the edges and connected to GND.

A butterfly arrangement (2×2 channels around a central header) was evaluated first and discarded: the ribbon cable would leave the centre of the board vertically, and the motor wiring would exit on two opposite sides. The final floorplan is **linear (1×4)**, with a single signal flow from the Jetson side to the motor side:

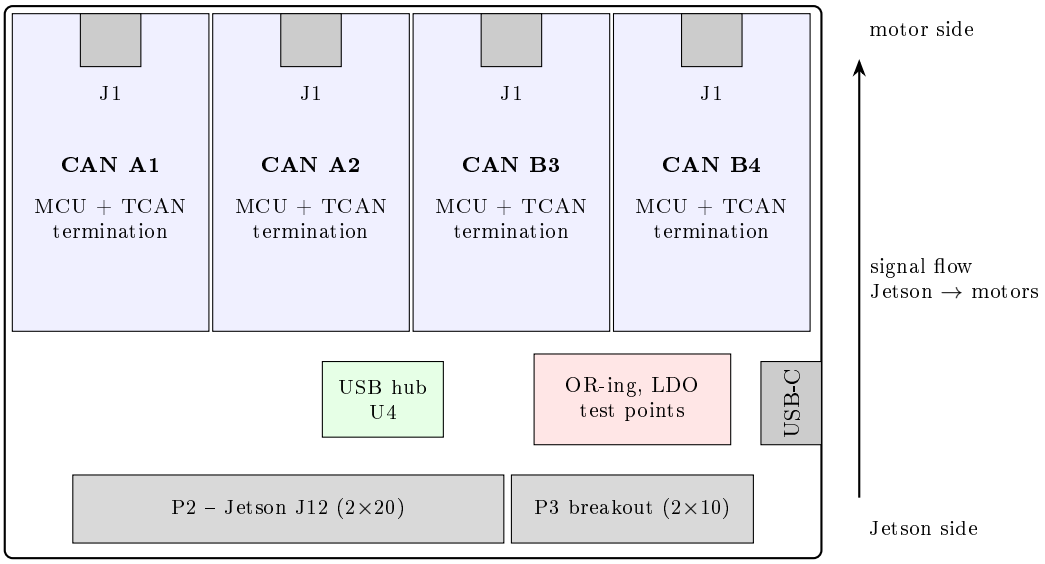


Figure 3: Floorplan (schematic, not to scale in detail). The four identical channels are placed in a row; the Jetson header, the breakout, the hub and the power stage share the bottom strip; the CAN connectors are on the top edge.

This arrangement keeps the clean Jetson/USB side and the noisy motor-cable side on opposite edges, lets all four channels have the same orientation, and allows the routing of one channel to be replicated on the others with *Copy Room Formats* without rotation.

9.3 Channel Placement

In every channel the MCU is oriented so that its pins face the parts they connect to: the SPI and USB pins face the bottom strip (P2 and the hub), and the CAN pins face the transceiver and the J1 connector. This keeps the tracks short and avoids crossings around the MCU:

MCU side	Pins	Signals
Bottom	25–36	SPI2 (25–28), USB PA11/PA12 (32, 33), SWDIO (34), VDDIO2 (36) – towards P2 and the hub
Right	37–48	SWCLK (37), BOOT0 (44), CAN PB8/PB9 (45, 46), VDD (48) – towards the transceiver
Top	1–12	VBAT (1), crystal PF0/PF1 (5, 6), NRST (7), VDDA (9), LEDs PA0/PA1 (10, 11)
Left	13–24	SPI_INT PB11 (22), VDD (24)

Table 9: Sides of the MCU in the chosen orientation.

Placement priorities around the MCU:

1. 100 nF decoupling capacitors at their VDD pins, with the via placed *after* the capacitor pad (pin → capacitor → via);
2. crystal Y1 directly in front of PF0/PF1 with C7/C8 beside it; no other track under or next to the oscillator. The two crystal tracks do not need to be equal in length: they are two different nodes of a Pierce oscillator, and a difference of a few millimetres adds only tenths of a picofarad;
3. C11/C12 on VDDA, then C4 on NRST (second row, NRST track leaving away from the crystal);

4. R10 next to BOOT0; R5 (STB pull-down) near the transceiver, its track kept away from CANH/CANL.

The TCAN1042 is placed to the right of the MCU, with pins 1–4 (TXD, GND, VCC, RXD) facing the MCU and pins 5–8 facing the bus side. The bus path is J1 → PESD1CAN → split termination → transceiver; the termination resistors are shunt elements and do not lengthen the signal path. CANH and CANL are routed around the 2×2 termination block and differ by 5.1 mm (24.1 vs 29.2 mm), i.e. about 19 ps, which is negligible against the 1 µs bit time.

9.4 Design Rules

Rule	Value
Clearance	0.15 mm (the 0.5 mm pitch of the LQFP48 and USB-C leaves 0.2 mm between pads)
Track width (default)	min 0.15, preferred 0.254, max 1 mm
Track width `3V3`, `GND`	min 0.2, preferred 0.3 mm
Track width `5V`, `5VUSB`, `5V_SPI`	min 0.3, preferred 0.6 mm
Vias	0.3 mm drill, 0.6 mm diameter, tented on both sides in the design and manufactured as plugged vias (JLCPCB no longer offers tented vias); minimum annular ring 0.1 mm
Differential pairs (USB)	width 0.25 mm, gap 0.15 mm (90.2 Ω differential, Figure 2), max uncoupled length 5 mm
Polygon connection	THT pads: relief (hand soldering); SMD pads and vias: direct
Solder mask sliver	0.1 mm; maximum hole size 3.5 mm (M3 holes)
Copper to board edge	0.3 mm (JLCPCB minimum 0.2 mm, 0.3 mm to 0.5 mm recommended for reliability)
Bottom GND pour clearance	0.3 mm (`InPolygon and OnLayer('Bottom Layer')`); the inner planes keep 0.15 mm so that via antipads do not merge into slots

Table 10: Main design rules (JLCPCB 4-layer capabilities with margin).

9.5 Signal Integrity Considerations

A track behaves as a transmission line only when it is longer than about one sixth of the distance travelled by a signal edge (6.5 ps/mm on FR4):

$$l_{\text{crit}} = \frac{t_r}{6 \cdot 6.5\,\text{ps/mm}} \tag{11}$$

What matters is the edge time of the *driver*, not the clock frequency. The SPI lines have two different drivers: MISO is driven by the STM32, configured at low output speed, while SCK, MOSI and NSS are driven by the TXB0108 translators of the Jetson carrier board, whose output edges are much faster:

Signal	Driver	Edge time	Critical length	Longest track
CAN, 1 Mbit/s	TCAN1042	20 ns to 50 ns	> 500 mm	≈ 30 mm
SPI MISO	STM32, low speed	≈ 12 ns	≈ 300 mm	< 112 mm
SPI SCK, MOSI, NSS	TXB0108 (Jetson)	0.6 ns to 2.5 ns	15 mm to 64 mm	112 mm

Table 11: Critical lengths. CAN and MISO are electrically short; the lines driven by the Jetson are transmission lines and were checked with a reflection simulation (SI report [17]).

An earlier version of this table assumed SPI edges of 5 ns to 10 ns for all lines (critical length above 130 mm). That assumption holds for MISO only; the simulation showed that it does not hold for the lines driven by the TXB0108. Controlled impedance is still not required for the SPI lines: their reflections are controlled with series terminations at the drivers (see below). Only the USB pairs are routed as 90 Ω differential pairs, which is good practice rather than a necessity at Full Speed (12 Mbit/s). The field solver gives 90.18 Ω for the routed geometry on both outer layers, well within the ±10 % tolerance (Figure 2).

SPI. SCK, MOSI and MISO of the two MCUs of a bus join at the pads of P2 (star topology). On SPI1 the pins are spread over the whole header (MOSI on pin 37, SCK on pin 13), so SPI1 MOSI is the longest line (112 mm). Its capacitive load (Jetson carrier, ribbon, PCB track and two MCU inputs, about 40 pF) is within the range recommended for the TXB0108 translators; widening the track would only increase it. SPI lines are spaced 1 mm centre-to-centre (≈3× the track width). Where NSS runs next to SCK for a few millimetres (at USB crossings) the coupled glitch is estimated at 10 mV to 15 mV, far below the ≈ 1 V input threshold. The SPI GPIOs (and SPI_INT) must be configured at low output speed in the firmware.

SPI simulation and SCK series resistors. The reflections of the SPI1 bus were simulated with the Altium Signal Integrity analyser on the routed board, using the IBIS models of the STM32F072 and of the TXB0108 (fastest driver model, i.e. worst case). The complete procedure, the models and all waveforms are in the separate SI report [17] (Section 3); the results are summarised here. Thresholds: STM32 $V_{IL}/V_{IH} = 0.99/2.31$ V, TXB0108 1.05/2.34 V.

Net	Worst receiver	Min. after rising edge	Result
MISO (R4 at the driver)	P2-22 (Jetson)	2.77 V	pass, margin 0.43 V
MOSI	MCU B4	1.60 V	re-enters the threshold band for ≈ 20 ns after the edge; acceptable for a data line, which is sampled 125 ns to 500 ns later
SCK, no resistor	MCU B4	2.09 V	marginal: 0.22 V below V_{IH} on the clock line
SCK, 33 Ω	MCU B4	2.79 V	**pass**, margin 0.48 V; peak reduced from 4.9 to 4.1 V

Table 12: SPI1 reflection results (SI report [17], Table 14). The SCK result with 33 Ω is the analysis of the final layout.

As a result of the simulation (SI report [17], Section 3.8 and Figure 4), a 33 Ω series resistor was added on each clock line: **R17 on SPI1_SCK (P2 pin 13) and R18 on SPI0_SCK (P2 pin 23)**. A source termination only works close to the driver, so both are placed on the top layer within 3 mm to 5 mm of the P2 pad and *before* the branch point of the star; the branch point is a via next to the resistor, from which the bottom-layer branch leaves. The part is the same as R4 (`C23140`, Basic), so no BOM line and no setup fee are added. The time constant 33 Ω × 40 pF ≈ 1.3 ns is negligible at 1 MHz to 4 MHz. SPI0 was not simulated: it has the same star topology at P2 and is treated by similarity. The simulation does not include the track of the Jetson carrier nor the ribbon cable, so SCK must be checked with an oscilloscope on the prototype with the final cable (Section 11).

USB. Five pairs are routed: J2 → U3 → U4 (upstream) and U4 → MCU (four downstream ports). The downstream pins of the SL2.1A (DM4, DP4 ... DM1, DP1 on pins 1–8) face the MCUs in mirrored order with respect to PA11/PA12, and D+/D− cannot be swapped in USB. Each pair is therefore swapped with one line passing under the other on the bottom layer (two

vias). For the channels whose pair also has to cross an SPI bundle, the swap is combined with the crossing: both lines go down, cross under the SPI lines, and come back up at two different points (four vias in total instead of six). The resulting skew of 1 mm to 2 mm ($\approx$ 13 ps) is negligible against the 83 ns bit time. The bottom-layer sections are kept over the continuous 3.3 V area of L3.

9.6 Power Distribution

- **L2** is a solid GND pour over the whole board, never split.
- **L3** is a 3.3 V pour. The 5 V supply of the four transceivers is routed inside it as a **tree** of 1 mm tracks starting at the OR-ing output, with one branch per transceiver. The branches are not joined at the top: a closed ring would isolate islands of the 3.3 V plane. Bottom-layer signals do not cross the 5 V tracks.
- Voltage drop on the longest branch (67 mm, 1 mm wide, 15.2 µm copper): $R = \rho L/(w\,t) \approx 76\,\mathrm{m\Omega}$, i.e. 5 mV to 11 mV at 70 mA to 140 mA. The transceivers therefore receive about 4.69 V even with a 4.75 V source, above the 4.5 V minimum.
- `5V_SPI` (P2 pins 2 and 4, joined) is routed on L3 with a width of at least 0.8 mm: according to IPC-2221 [15] an inner 0.5 oz track needs about 0.7 mm for 0.5 A with a 10 °C temperature rise. Layer transitions use two or three vias (about 1 A each).
- Small top-layer pours connect the OR-ing (`5V_SPI`, `5VUSB`, `5V` kept as separate nets) and the AMS1117 output (tab, pin 2, C21, TP1), with thermal vias to the 3.3 V plane. Small pours are poured before the large planes (pour order), so that the planes flow around them.
- `3V3_Jetson` is a separate net routed only as a track to P3; it never touches the 3.3 V plane.
- **L4** carries a GND pour over the whole board, stitched to L2 with GND vias (about every 5 mm to 10 mm, along the edges, at the end of narrow copper fingers and next to the signal vias of the bottom-layer crossings). It gives the bottom-layer tracks a nearby GND return and improves the copper balance. No pour was added on the top layer: there the free copper is fragmented between components and would create many small, poorly stitched areas, with little benefit over the solid GND plane on L2.

The 3.3 V pour on L3 was checked after all pours and stitching vias: it remains a single connected area (no island is isolated by the 5 V tree or by via antipads). Since L3 is a signal-type layer with a polygon, Altium's automatic via stitching is blocked by the 3.3 V polygon; the stitching vias were therefore placed manually.

10 Conclusion and Recommendations

The BYTE SPIne provides four independent CAN buses for the twelve actuators of the BARK quadruped, with one MCU and one transceiver per leg. The same hardware supports a ready-to-use USB/candleLight mode and the final SPI mode, so development can start immediately without a second board revision.

The key sizing choices are:

- an external 8 MHz crystal per MCU, required for CAN timing without USB;
- a split termination built from 2 $\times$ 120 Ω 1206 per half, sized for the 100 mA fault current of the transceiver;

- an ideal-diode OR-ing (LM66100, figure 17 configuration) that keeps the 5 V rail above the 4.5 V minimum of the TCAN1042;
- a 100 kΩ NSS pull-up, compatible with the TXB0108 translators of the Jetson;
- 33 Ω source terminations on MISO (R4) and, as a result of the signal integrity simulation, on both SCK lines (R17, R18).

The reliability of the SPI mode depends mainly on the firmware.

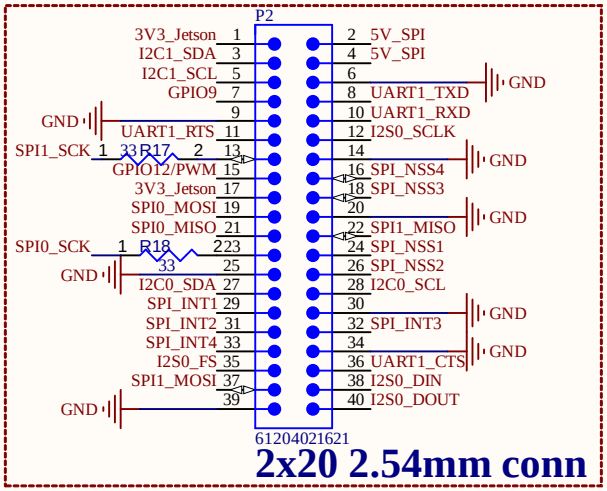
Required: implement the MCU-side CAN loop, the **CRC and sequence counter** on SPI packets and the **communication watchdog** before the robot is operated in SPI mode.

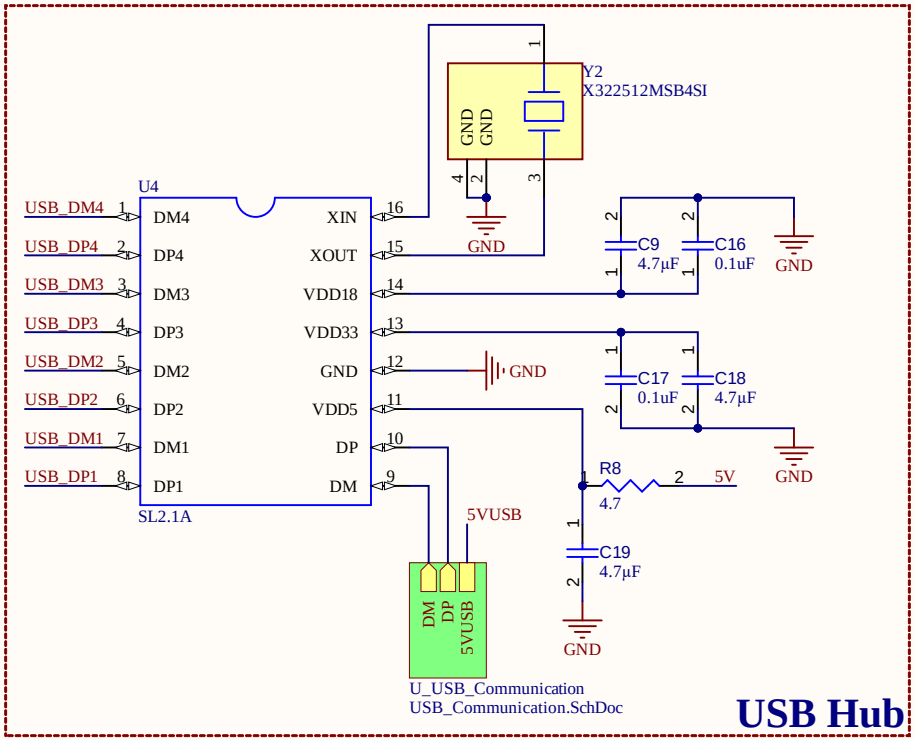
11 Next Steps for future iterations

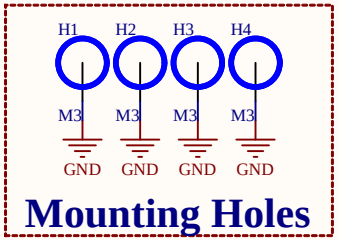
- Bench validation of the prototype (version 1.0):
  - stability of the 3.3 V rail with the ceramic output capacitor;
  - 5 V rail voltage under full CAN load, from both sources;
  - MISO release with two MCUs on the same SPI bus;
  - SCK at both MCUs of a bus with an oscilloscope (short ground spring), with the final ribbon cable, to confirm the simulation;
  - SPI link at 1 kHz for 10 minutes with zero lost packets (sequence counter and CRC);
  - pinout of the CAN cables (`H L G`) against the SteadyWin connector.
- Remaining simulations (SI report [17], Section 5): SCK with the realistic TXB0108 model, NSS falling edge, crosstalk SCK → NSS, SPI0 on the layout.
- Version 1.1: fixes resulting from the bench tests; schematic clean-up (electrical type of the supply pins of the TCAN1042 and SL2.1A symbols, one single name for the `5VUSB` net).
- Optional improvement: a right-angle version of P2 would let the ribbon cable leave parallel to the board.
- Firmware: candleLight build for the BYTE SPIne pinout; SPI slave firmware with watchdog.
- Verify the termination of the actuators (only the last motor of each leg terminated).

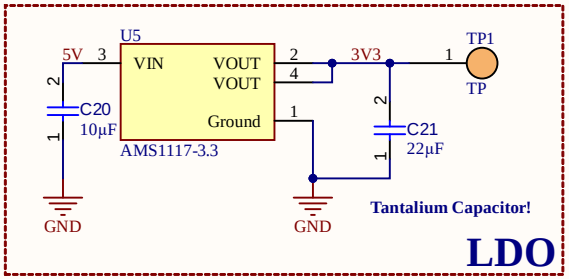
12 Annexes

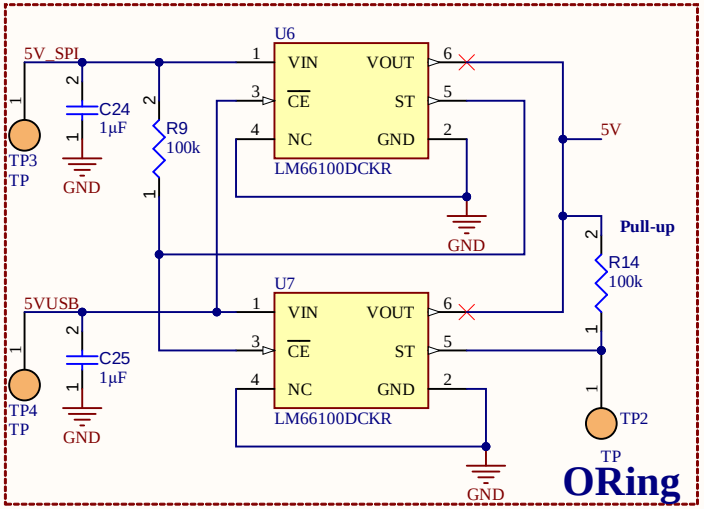
Complete schematic of the BYTE SPIne (`BYTE_SPIne_Schematic.pdf`, generated by the Output Job of the project): top sheet, one CAN channel (instantiated four times) and USB hub.

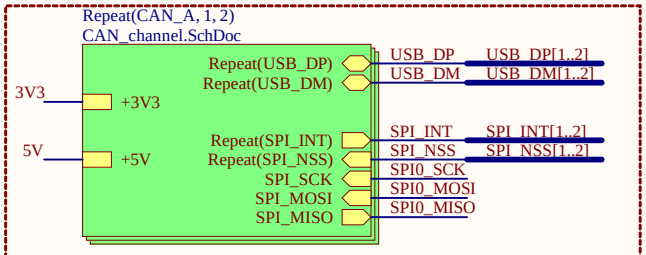


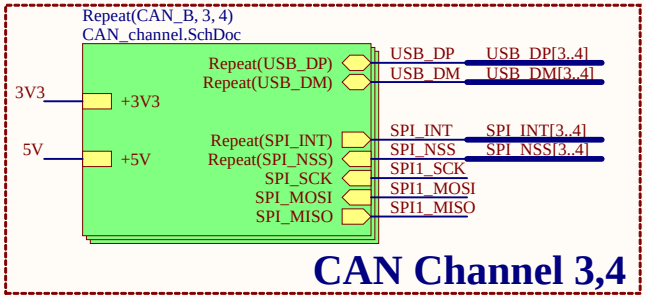


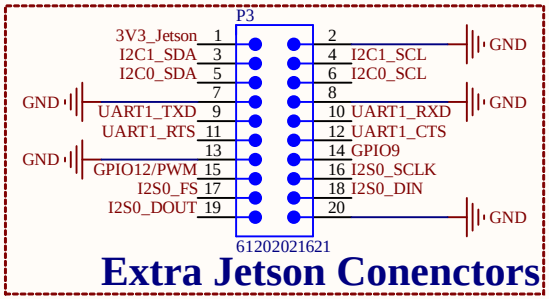




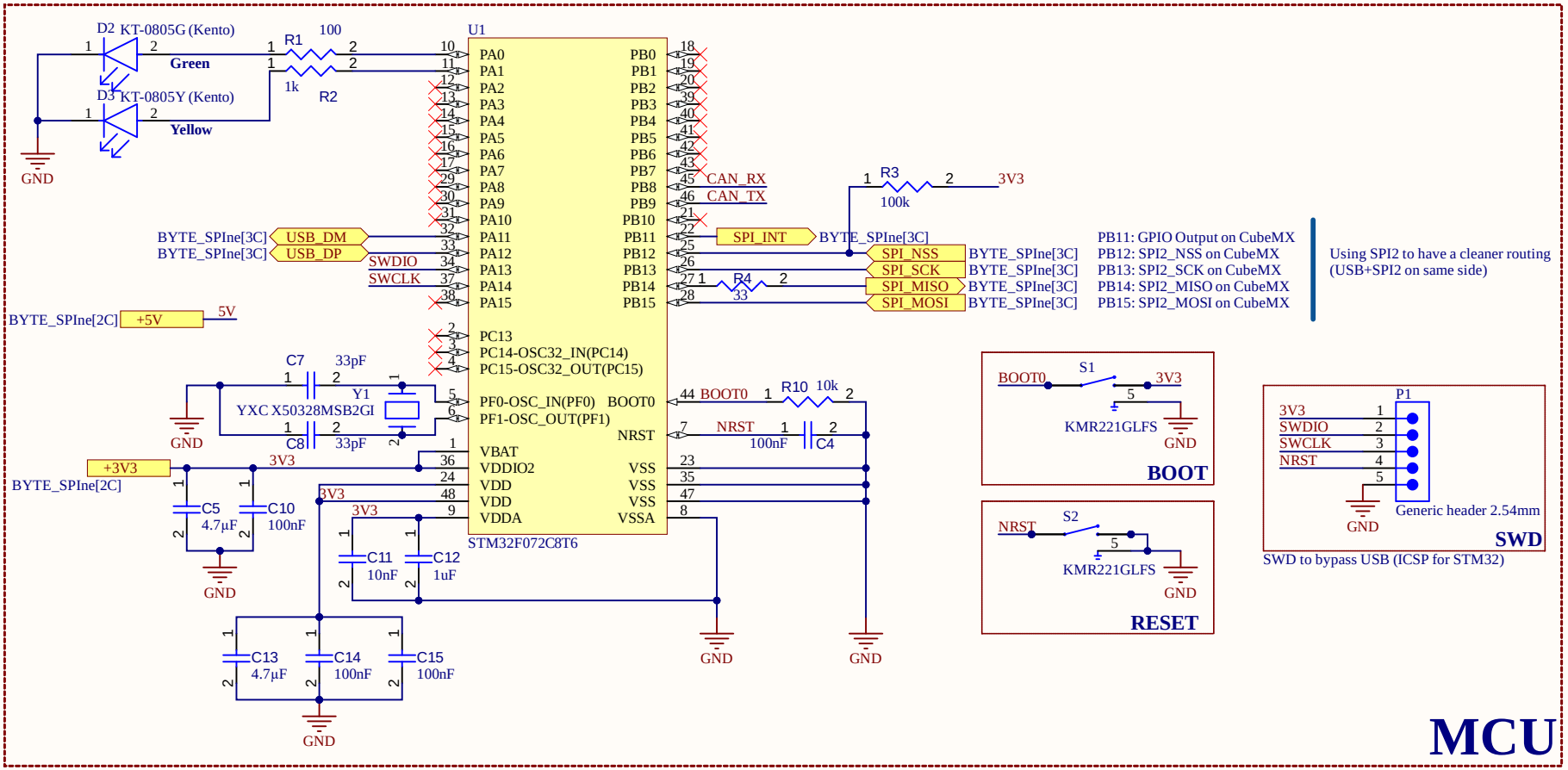


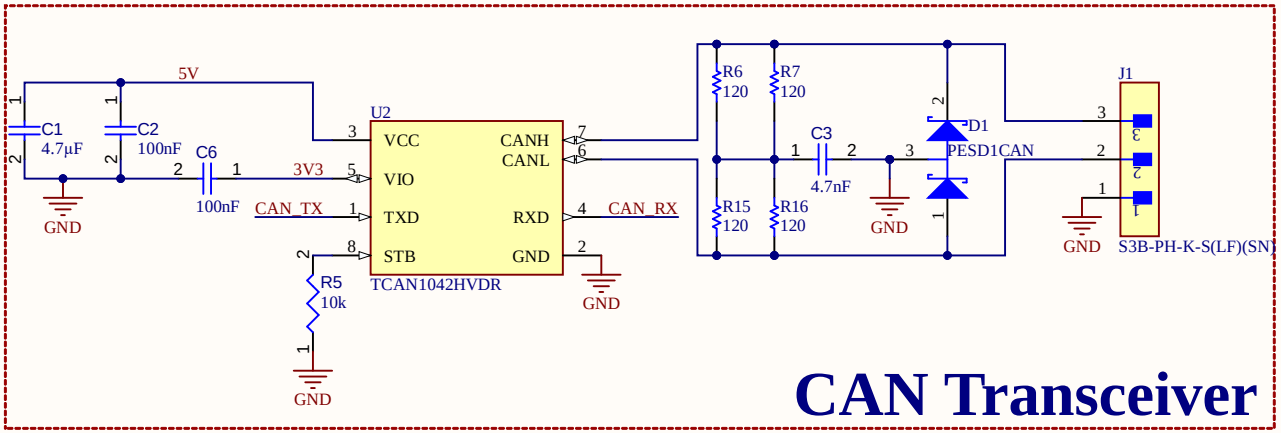




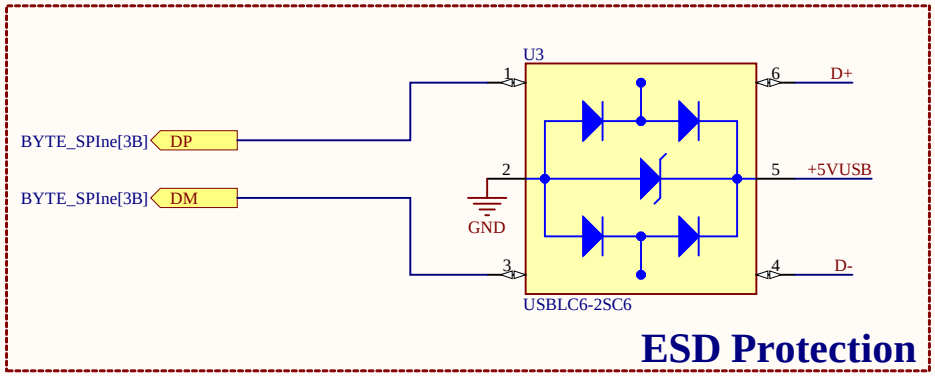


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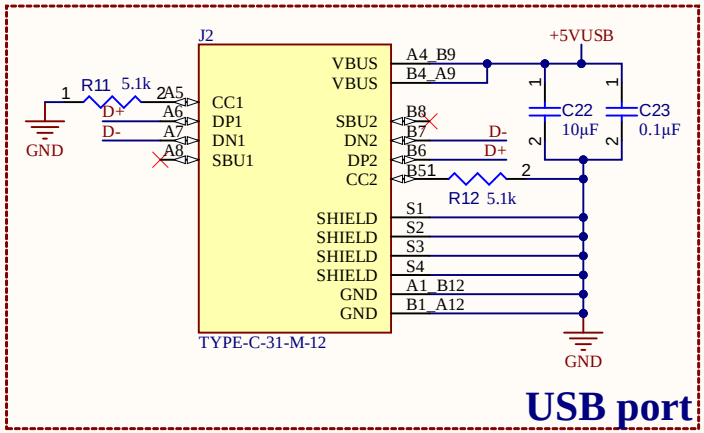




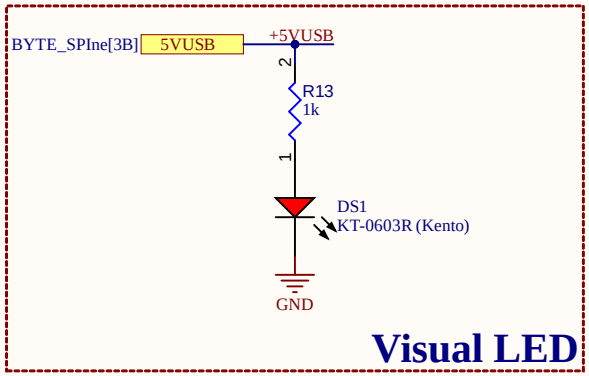
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ESD Protection



USB port



Visual LED

Title: **BYTE SPIne - USB-C & 4-Port Hub**		
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Date: 10.29.2026		Sheet 3 of 3
File: C:\Users\..\USB_Communication.SchDoc		Drawn By: Johnan Luca

13 References

[1] STMicroelectronics, *STM32F072x8/xB datasheet*, DocID025004. Product page with datasheet, reference manual and application notes: https://www.st.com/en/microcontrollers-microprocessors/stm32f072c8.html.

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